

Gain and absorption characteristics of bilayer quantum dot lasers beyond 1.3 μm

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ABSTRACT

In this paper we report on the multi-section gain and absorption analysis of strain engineered molecular beam epitaxy (MBE) grown GaAs and InGaAs capped bilayers. The InGaAs capped bilayer quantum dot (QD) lasers extends the room temperature lasing wavelength to 1.45 μm . The spectral measurement of gain demonstrates that net modal gain is achieved beyond 1.5 μm at room temperature. Analysis of the temperature and current density dependence gain characteristics of a GaAs capped bilayer sample indicate that the temperature sensitivity of threshold current around room temperature is due to phonon assisted thermal escape of carriers from the QDs.

Keywords: Bilayer, Quantum dots, Multi-section devices, Gain and Absorption

1. INTRODUCTION

GaAs based Quantum dot (QD) materials are commercialized in the 1.0 μm to 1.3 μm region. However, many of the properties of QD lasers (zero chirp, temperature insensitive operation, single photon emission) are particularly attractive at 1.55 μm . If possible GaAs based optoelectronics would then be able to service all the major optical communications wavelengths from red to the C-band in the infrared. In addition, excited state lasing at 1.3 μm is attractive for high modulation rates¹⁻³. The use of bilayers of QDs has been demonstrated as a possible route to obtaining long wavelength emission and lasing⁴⁻⁷. In a bilayer structure two strain and electronically coupled QD layers are grown instead of a single InGaAs capped QD layer. The concept is that the lower lying QD layer controls the QD density of the upper layer, and the growth conditions of the upper QD layer can determine the emission wavelength. Recently, we have demonstrated room temperature electroluminescence up to 1.52 μm ⁸ from devices incorporating quantum dot bilayers without the need for an InGaAs metamorphic buffer layer⁹. These structures are promising for the realization of GaAs-based QD devices in the 1.55 μm region. In this paper we compare experimental net modal gain and modal absorption spectra at a range of current and temperature using the modified variable stripe length method in multi-section devices¹⁰. These results are useful for understanding fundamental process in QDs such as increase in threshold current density with temperature and key to optimizing distributed feedback lasers (DFBs) and vertical cavity surface emitting lasers (VCSELs) that generally have larger losses than Fabry-perot devices. Under high bias conditions, asymmetric broadening of the gain peaks towards longer wavelengths is observed, which results in positive net modal gain beyond 1.5 μm for a laser incorporating InGaAs-capped QD bilayers.

2. DEVICE DESCRIPTION AND EXPERIMENTAL DETAILS

The laser structures described here are shown schematically in Fig. 1. Growth was carried out on n⁺ GaAs (100) substrates by molecular-beam epitaxy (MBE). The active region for both structures was located in a 500nm undoped GaAs layer sandwiched between 1500nm Al_{0.33}Ga_{0.67}As cladding layers. A 400 nm p-type GaAs:Be contact layer was grown to complete the laser structure. The active region of the 'standard' structure consists of five GaAs capped QD bilayers. We refer to this sample as the "GaAs capped bilayer" sample in the following.

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Each bilayer consists of a seed layer of InAs/GaAs QDs and an emission layer. Before each QD layer was grown, the surface was annealed under an As₂ flux at 580°C for 10 minutes to smooth the growth surface⁴. Thereafter the seed layer was grown at a temperature of 480°C by the deposition of 2.4 ML of InAs at a growth rate of 0.014 MLs⁻¹, giving a QD density of 2.7 x 10¹⁰ cm⁻². These seed QDs were capped by 10nm of GaAs spacer layer, also grown at 480°C, before the temperature was raised again to 580°C for 10 minutes to smooth the growth surface and desorb segregated indium.

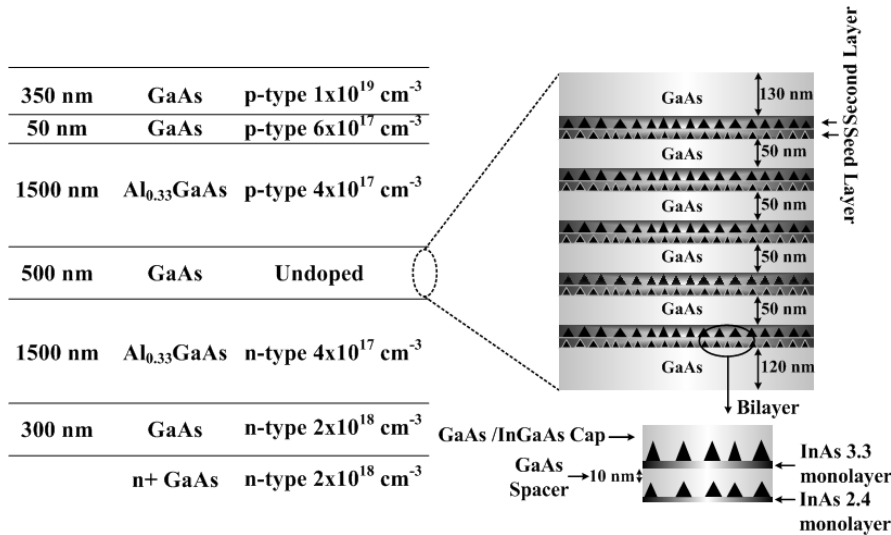


Figure 1. Schematic diagram of bilayer QD laser structures.

The emission layer was then grown at a reduced temperature of 467°C by the deposition of 3.3 ML of InAs at the same growth rate as for the seed layer. The QDs were then capped with 15 nm of GaAs also at 467°C, after which the temperature was ramped to 580°C for subsequent GaAs growth. The same sequence was repeated for the remaining of the active region with the remainder of the structure grown at 580°C. The second structure discussed is identical to the GaAs capped bilayer sample except the upper emission layer (second QD layer) is capped by 4nm of In_{0.18}Ga_{0.82}As before subsequent GaAs growth. We term this as the “InGaAs capped bilayer”. Wafers were processed into broad area lasers (15 to 100 μm ridge width) for multilength analysis and 7 μm ridge width multi-section devices for variable stripe analysis¹¹. First, the characteristics were measured in the pulsed regime (5 μs pulsed duration, 1% duty cycle) to minimize thermal effects and then measured in the CW regime to assess performance under realistic operating conditions. All measurements were performed at a tile temperature of 298K.

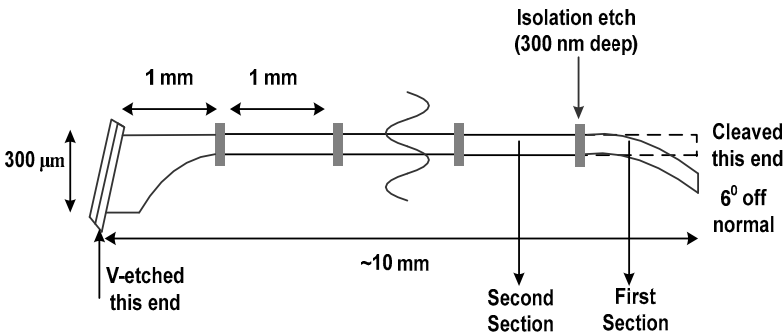


Figure 2. Schematic of a multi-section device.

Fig. 2 shows the schematic of the multi-section device. It consists of 9 mm long, 7 μm wide ridges with 1mm isolated contact sections. At the rear of the device is a ~1 mm long, 300 μm wide tapered absorber section with a tilted deep V-etched back facet, to suppress round trip amplification for single pass measurements. For some of the devices, the front

(output) facets are either perpendicular to the cleavage plane or at 6° off to the axis of the waveguide. The front curvature was modeled to give minimum loss at wavelengths >1000 nm. Laser structures with and without isolated sections were fabricated to allow the characterization of both the material and fabrication process. Further details of the fabrication process can be found in ref ¹¹. The segmented contact method was employed to obtain gain and absorption spectra¹⁰. To obtain gain, single pass amplified spontaneous emission (ASE) intensity was captured by a single mode optical fibre and measured using an optical spectrum analyzer. The ASE intensity resulting from pumping the first section (see Fig. 2) of length L (I_L) and from pumping the first two sections of length 2L (I_{2L}) was recorded at various current densities. The net modal gain, G, was obtained from:

$$G = \frac{1}{L} \left[\ln \left(\frac{I_{2L}}{I_L} - 1 \right) \right] \quad (1)$$

The optical loss was found by measuring the ASE from pumping the first section (I_1) and then pumping only the second section (I_2) under the same current density while converting the first section ($V_{\text{applied}} = 0$ V) into an absorber of length L. As long as the spontaneous emission from both section are equal the optical loss suffered in section 1 is given by:

$$\alpha = \frac{1}{L} \left[\ln \left(\frac{I_2}{I_1} \right) \right] \quad (2)$$

3. RESULTS AND DISCUSSION

Fig. 3, shows the example of the measured ASE at room-temperature on a bilayer laser structure while pumping section 1 (length L=1mm), then pumping section 1 and 2 (total length 2L), for a drive current density of 3 kA/cm² (pulse regime).

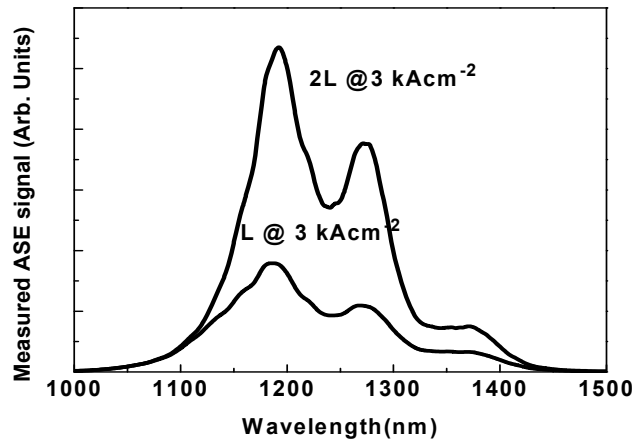


Figure 3. Measured ASE spectra for length L and 2L at 3 kA/cm².

The net modal gain spectra of two different bilayer devices with QDs capped by either GaAs or In_{0.18}Ga_{0.82}As are shown in Fig. 4. In this technique, by changing the pumping current a series of net modal gain spectra can be obtained, but for comparison we have selected current density of 3 kA/cm², where the ground state (GS) and first excited state (ES1) is clearly resolved for both the devices. At this current density for the GaAs cap bilayer sample we observe the GS (1.37 μm) and ES1 (1.27 μm) net modal gain of 6 cm⁻¹ and 13 cm⁻¹ respectively. Similarly, for the InGaAs cap GS¹ (1.45 μm) and ES¹ (1.33 μm) net modal gain of 3 and 5 cm⁻¹ are observed. The gain peak wavelength is in good agreement with

emission spectra obtained from the laser devices (not shown). QD density and/or increased dot layer number can be expected to increase the net-modal gain significantly for high speed applications². Similarly, operation of the excited

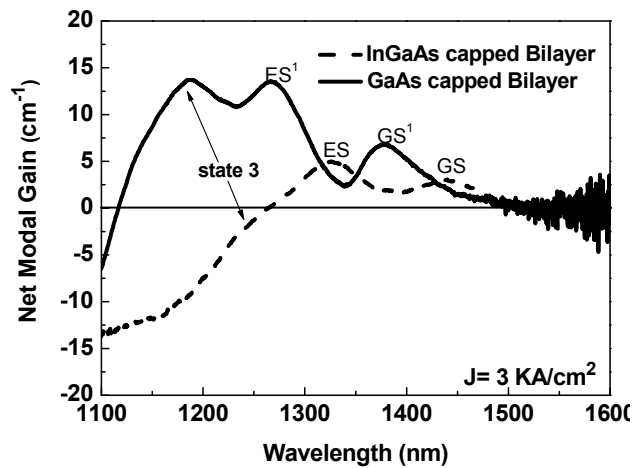


Figure 4. Net modal gain spectra of InGaAs and GaAs capped bilayer device as measured using the multi-section technique.

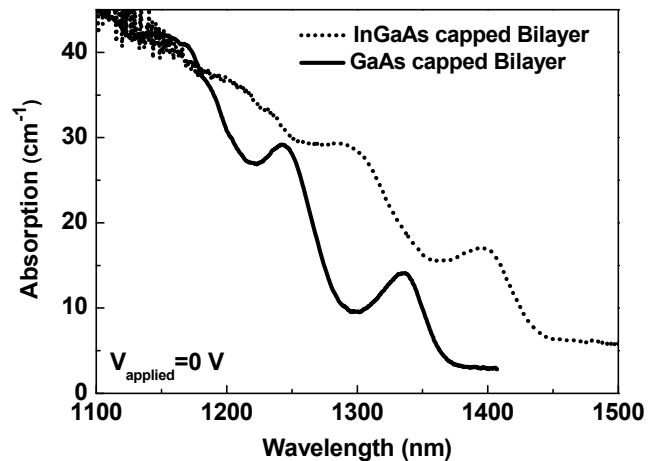


Figure 5. Modal Absorption spectra of InGaAs and GaAs capped bilayer device as measured the using multi-section technique.

state at 1.3 μ m offers increased modulation bandwidth³. Significantly, we observe a positive net modal gain beyond 1.5 μ m for the InGaAs capped bilayer device. This is a step towards achieving telecoms devices on GaAs substrates.

Fig. 5 shows the modal absorption spectra of GaAs and InGaAs cap bilayer devices, respectively, plotted for a current density where no bleaching in the peak absorption is observed. The absorption peaks arising from the ground state and the first excited state of both samples are clearly observed, and the shift in the ground state absorption peak between GaAs capped and InGaAs capped bilayer indicates that these transitions must be related to the upper QD layer. The absorption does not vary with wavelength below the band-edge of the material indicating an internal loss $\alpha_i = 3.0 \pm 0.5 \text{ cm}^{-1}$ for sample A and $\alpha_i = 6.0 \pm 0.5 \text{ cm}^{-1}$ for sample B. The increased internal loss for InGaAs capped bilayer is in line with the increased roughness of the lower AlGaAs/GaAs interface observed in TEM images (not shown). Subtracting α_i for each sample gives a very similar ground state peak absorption of 11 cm^{-1} for both samples; as the QD densities are nominally the same in both samples this also indicates that there are a similar number of optically active QDs in each sample and this is not affected by the introduction of the InGaAs capping. This indicates that in the bilayer,

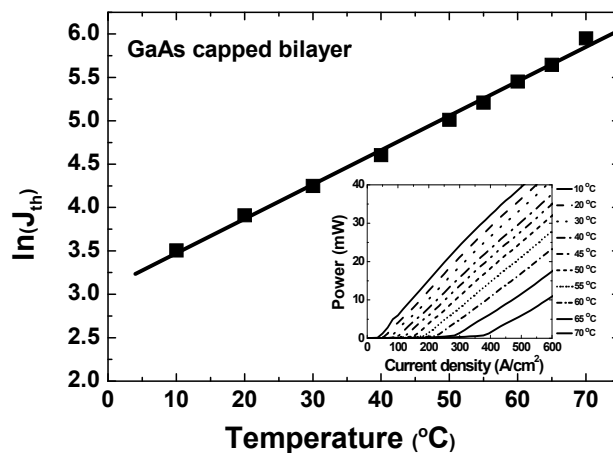


Figure 6. Natural logarithm of threshold current versus temperature of 4mm long and 15 μ m wide GaAs capped bilayer laser. Inset shows the Power-current characteristics for a range of temperatures.

the QDs of the seed and emission layers have not just strong correlation structurally but also electrically⁵. Interestingly

we noted for GaAs capped bilayer (InGaAs capped bilayer), in the absorption spectrum the ground state and first excited state are observed at $\sim 1.35 \mu\text{m}$ ($\sim 1.4 \mu\text{m}$), and $\sim 1.25 \mu\text{m}$ ($\sim 1.3 \mu\text{m}$). The corresponding transitions in the gain spectrum are observed at $\sim 1.37 \mu\text{m}$ ($1.45 \mu\text{ms}$) and $\sim 1.26 \mu\text{m}$ ($1.33 \mu\text{ms}$). This shift between the absorption and gain spectrum of quantum dots can be attributed to the Coulomb interaction between the QDs and neighboring charge¹². Higher order transitions are also observed. The assignment of these transitions is difficult in this coupled QD system at high dot carrier occupancies, and in the remainder, the peak observed at $1.15 \mu\text{m}$ ($1.21 \mu\text{m}$) in absorption ($1.18 \mu\text{m}$ ($1.23 \mu\text{m}$) in the gain spectrum) is referred to as “state 3” for GaAs capped sample (InGaAs capped). We note that the values for absorption at the ground state and first excited state peaks are significantly larger than the corresponding gain^{10, 13}.

In the remainder of this report we focus on the GaAs capped bilayer structure. Fig. 6 shows natural logarithm of threshold current density of the 4mm long and $15\mu\text{m}$ wide GaAs capped bilayer laser versus temperature. The inset shows the power-current characteristics at a range of temperatures. The threshold current density increases with increase in temperature from 10-70°C. Insight into the origin of the change of threshold current in this important temperature range can be assessed by analyzing the gain spectra. In addition optical gain will allow us to determine spectral width and modulation behavior at a range of temperature which is critical for telecommunications applications.

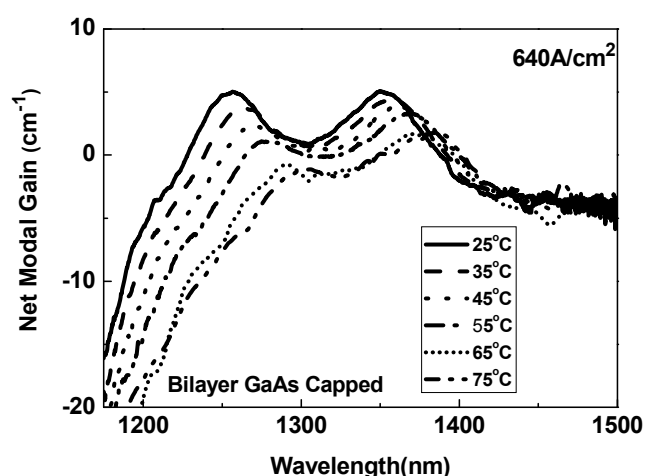


Figure 7. Net modal gain spectra versus temperature of GaAs capped bilayer device at 640 A/cm^2 as measured using the multi-section technique.

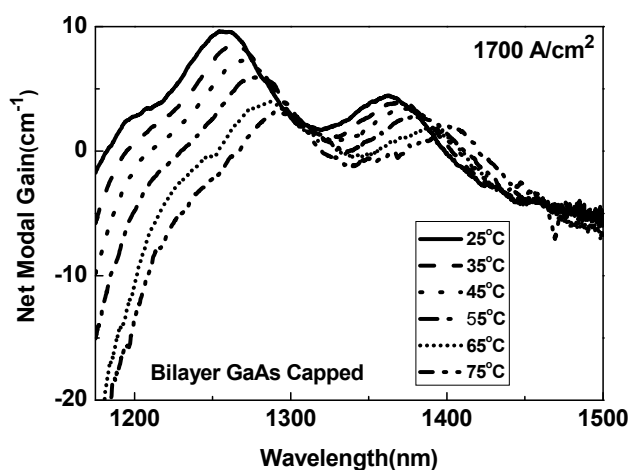


Figure 8. Net modal gain spectra versus temperature of GaAs capped bilayer device at 1700 A/cm^2 as measured using the multi-section technique.

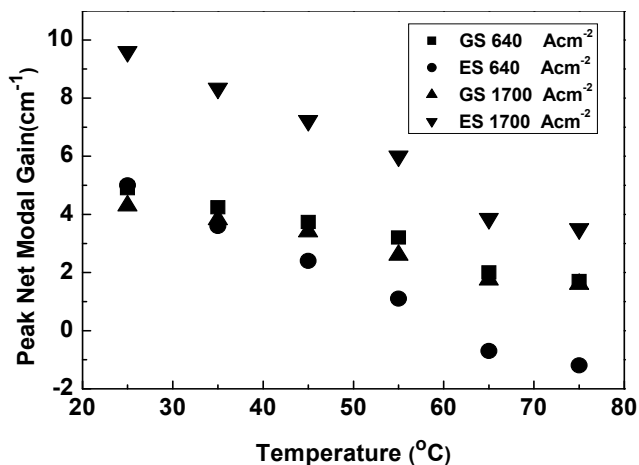


Figure 9. Peak net modal gain versus temperature of GaAs capped bilayer device at 640 and 1700 A/cm^2 as measured using the multi-section technique.

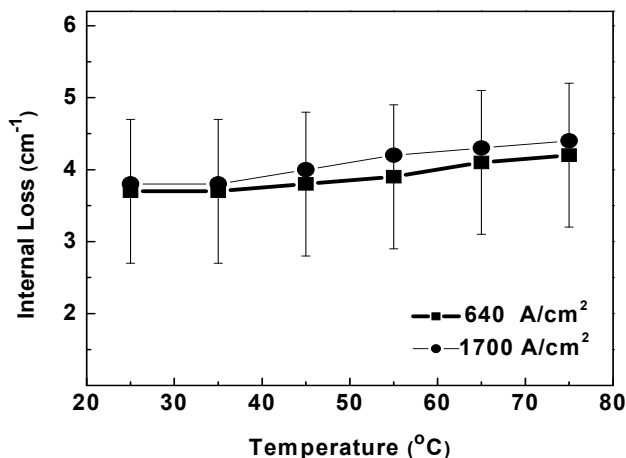


Figure 10. Internal loss versus temperature of GaAs capped bilayer device at 640 and 1700 A/cm^2 as measured using the multi-section technique.

Fig. 7 and 8, shows the net modal gain spectra versus temperature of the GaAs capped bilayer device at 640 A/cm^2 and 1700 A/cm^2 (CW) as measured using the multi-section technique. We have chosen to plot the net modal gain versus

temperature at two specific points first at 640A/cm², where the ground state is fully saturated and excited state is at a similar peak gain. And second points at 1700A/cm² where the ground and excited state are both fully saturated and the excited state peak gain is double the peak gain in ground state. These two conditions correspond to quite different dot occupancies, yet in both cases the GS gain is saturated.

Figs.9 & 10 shows the peak net modal gain and internal loss plotted against temperature for the GaAs capped bilayer device derived from Figs. 7 & 8. Interestingly the ground state saturated gain has the same dependence on temperature at 640 and 1700A/cm². We note a steeper gain reduction in excited state indicating faster carrier escape (carrier density as a whole reducing). Whilst the ground state is saturated at both current densities but peak gain reduces with temperature (with variation (± 0.5) in internal loss), indicating non-radiative recombination from carriers after reaching the ground state. It would be expected that at higher current densities (1700A/cm²) a larger reduction in gain would be observed if carrier-carrier effects were key in reducing gain. Furthermore, the excited state is observed to have a saturated gain of double the gain of the ground state at all the temperatures. This data suggest that the increase in threshold current density with temperature is associated with phonon assisted thermal escape of carriers from the dots.

4. CONCLUSION

We have shown a method of extending the emission wavelength of QDs on GaAs by using a bilayer growth technique. InGaAs capping the bilayer of QDs further extends the emission wavelength to 1.45 μ m. By optimization of the QD capping material it is possible to achieve long wavelength emission without sacrificing modal gain. Indeed, net modal gain above 1.5 μ m is noted, presenting an exciting opportunity towards fabricating optoelectronic devices with all the advantages of QDs in the telecoms windows of 1.3 and 1.55 μ m.

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